

1. Administrative & Study Identification

Full Study Title: A Phase I/II, Open-Label, Multicenter, Study Evaluating the Safety, Pharmacokinetics, and Efficacy of Alectinib in Pediatric Participants With ALK Fusion-Positive Solid or CNS Tumors for Whom Prior Treatment Has Proven to be Ineffective or for Whom There is No Satisfactory Treatment Available **Short Title/Acronym:** A Study Evaluating the Safety, Pharmacokinetics, and Efficacy of Alectinib in Pediatric Participants With ALK Fusion-Positive Solid or CNS Tumors (iMATRIX Alectinib) **Protocol Number:** G042286 **NCT Number:** NCT04774718 **Trial Status:** Recruiting **Sponsor Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Alectinib / Alecensa (oral capsule/suspension) **Phase of Development:** Phase 1 / Phase 2 **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, pharmacokinetics, and efficacy of alectinib in pediatric participants, to assess the incidence of dose-limiting toxicities (DLTs), adverse events, and alectinib plasma concentrations. **Secondary Objectives:** To determine Confirmed Objective Response Rate (ORR), Duration of Response (DOR), and Time to Response (TTR).

2.2 Background & Rationale Alectinib is a central nervous system (CNS)-penetrant, oral, next-generation inhibitor of ALK-fusion proteins that has demonstrated durable responses in adults. In children and adolescents, there remains a significant unmet clinical need for patients with tumors harboring ALK-fusions (e.g., infantile high-grade gliomas, inflammatory myofibroblastic tumors) at diagnosis or relapse, for whom prior treatment has proven ineffective or for whom there is no satisfactory standard treatment available.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 42 participants **Number of Sites:** Multicenter (Global, including USA, UK, France, Germany, Canada, Australia, Brazil, China, etc.) **Estimated Study Start:** [DD-MMM-YYYY] **Estimated Primary Completion:** [DD-MMM-YYYY]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed diagnosis of CNS or solid tumors with documented evidence of ALK gene fusions as assessed centrally through the use of the investigational F1CDx assay or based on pre-existing NGS test results. 2. Disease status: prior treatment proven to be ineffective (i.e. relapsed or refractory), or for whom there is no satisfactory standard treatment available. Disease should be measurable and evaluable as defined by Response Evaluation Criteria in Solid Tumors (RECIST) v 1.1, or Response Assessment in Neuro-oncology criteria (RANO) +/- bone marrow criteria for primary CNS tumors or International Neuroblastoma Response Criteria (INRC). 3. Available tumor tissue for submission from active disease or archival tumor tissue, or willingness to undergo a biopsy sample collection. 4. Lansky Performance Status $\geq 50\%$ (for < 16 years) or Karnofsky Performance Status $\geq 50\%$ (for ≥ 16 years). 5. Adequate bone marrow function within at least 28 days prior to initiation of study drug. 6. Participant and/or caregiver willingness and ability to complete clinical outcome assessments throughout the study. 7. Agreement to remain abstinent or use contraception for males and females of childbearing potential, and agreement to refrain from donating eggs or sperm.

4.2 Exclusion Criteria 1. Medical history of: prior use of ALK inhibitors; diagnosis of Anaplastic Large Cell Lymphoma (ALCL); gastrointestinal disorders that may affect absorption; history of organ transplant; stem cell infusions. 2. Substance abuse within 12 months prior to screening. 3. Familial or personal history of congenital bone disorders, bone metabolism alterations, or osteopenia. 4. Treatment with investigational therapy 28 days prior to initiation of study drug. 5. Liver or kidney disease. 6. CTCAE v5.0 grade ≥ 3 toxicities attributed to any prior therapy which have not shown improvement. 7. Co-administration of anti-cancer therapies other than those administered in this study. 8. Active hepatitis B or C virus (HBV, HCV), or known HIV-positivity or AIDS-related illness. 9. Any clinically significant concomitant disease, psychological, familial, sociological, or geographical condition potentially hampering compliance, or planned procedure/surgery. 10. Infection considered clinically uncontrolled or of unacceptable risk upon induction of neutropenia. 11. Pregnant or breastfeeding, or intending to become pregnant during the study or within 3 months after the final dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Alectinib administered twice a day (bd) according to initial body weight in 28-day cycles. **Route of Administration:** Oral (tablet or

suspension). **Duration of Treatment:** Until disease progression or unacceptable toxicity (evaluated up to 10 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive pediatric oncology care. **Prohibited:** Other investigational drugs, prior ALK inhibitors, strong CYP3A inhibitors/inducers, and other anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor tissue/biopsy collection, ALK fusion confirmation (NGS/F1CDx), baseline scans
Baseline	Day 0	Enrolment, First Dose of Alectinib
Interim	Cycle 1 to X	Ongoing 28-day cycles: DLT assessment (Cycle 1), PK sampling, safety labs, efficacy scans (RECIST/RANO/INRC)
End of Study	Up to 10 Yrs	Final Vital Signs, Exit Interview, IP Accountability, Disease progression documentation

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Incidence of Participants with Dose-Limited Toxicities (DLTs). 2. Percentage of Participants with Adverse Events. 3. Plasma Concentration of Alectinib.

7.2 Secondary Outcomes 1. Confirmed Objective Response Rate (ORR) as determined by the Investigator. 2. Duration of Response (DOR) as determined by BICR and the Investigator. 3. Time to Response (TTR) as determined by BICR and the Investigator.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring up to 10 years to determine the percentage of participants with AEs. Key events monitored include CPK, AST, ALT, bilirubin increases, neutropenia, encephalopathy, and weight gain. **Serious Adverse Events (SAEs):** Investigated for relationship to Alectinib, reported within standard regulatory timelines. **Data Monitoring Committee (DMC):** Internal/External safety review to confirm the Recommended Phase 2 Dose (RP2D) during the Phase 1 part of the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety-evaluable population (patients receiving ≥ 1 dose of Alectinib); Efficacy-evaluable population (patients with measurable disease who received ≥ 1 dose of Alectinib).

dose). **Sample Size Justification:** Exploratory pediatric basket trial with a target enrollment of 42 participants to define RP2D and observe initial efficacy signals. **Handling of Missing Data:** Standard censoring methods for time-to-event endpoints (DOR, PFS, OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular clinical monitoring of dosing and AE tracking, coupled with Blinded Independent Central Review (BICR) for efficacy endpoints. **Audit Trail:** Standard tracking of eCRFs and central laboratory/imaging queries.

11. Study Identifiers & Governance

Primary ID: {{G042286}} **Registry Number:** {{NCT04774718}} **Posting ID:** {{34040583456907026}} **Sponsor/Lead Organization:** {{Roche}} **Study Title:** {{A Study Evaluating the Safety, Pharmacokinetics, and Efficacy of Alectinib in Pediatric Participants With ALK Fusion-Positive Solid or CNS Tumors}} **Official Regulatory Title:** {{A Phase I/II, Open-Label, Multicenter, Study Evaluating the Safety, Pharmacokinetics, and Efficacy of Alectinib in Pediatric Participants With ALK Fusion-Positive Solid or CNS Tumors for Whom Prior Treatment Has Proven to be Ineffective or for Whom There is No Satisfactory Treatment Available}}

12. Clinical Framework (Specific)

Primary Condition: {{ALK Fusion-positive Solid or CNS Tumors (Central Nervous System Neoplasms / ALK Fusion-Positive Tumor)}}

Diagnosis Threshold: {{ALK gene fusion positive via F1CDx assay or NGS}}

Medical Methodology: {{Targeted ALK Tyrosine Kinase Inhibitor Therapy}}

Optimization Target: {{Safety (DLT/AE profiles), Pharmacokinetics, and Tumor response (ORR/DOR)}}

13. Study Procedures & Methodology

Management Pathway: {{Treatment of refractory/relapsed pediatric tumors}} **Education Protocol (HCP):** {{Protocol training and specific DLT/AE management}}

Education Protocol (Patient): {{Informed consent, clinical outcome assessment (COA) completion, caregiver education on oral suspension/capsule administration}}

Quality Audit Mechanism: {{Blinded Independent Central Review (BICR) for scan evaluations}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 10 years}} **Onsite Visit Cadence:** {{Every 28-day cycle}} **Remote Monitoring Frequency:** {{Standard interval per site protocol}} **Checklist Review Interval:** {{Ongoing AE and DLT monitoring}}

15. Sign-off & Approval

Role	Name	Signature	Date			:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]						